

Propagation as Data (PaD): Neural Phase Hologram Generation with Variable Distance Support

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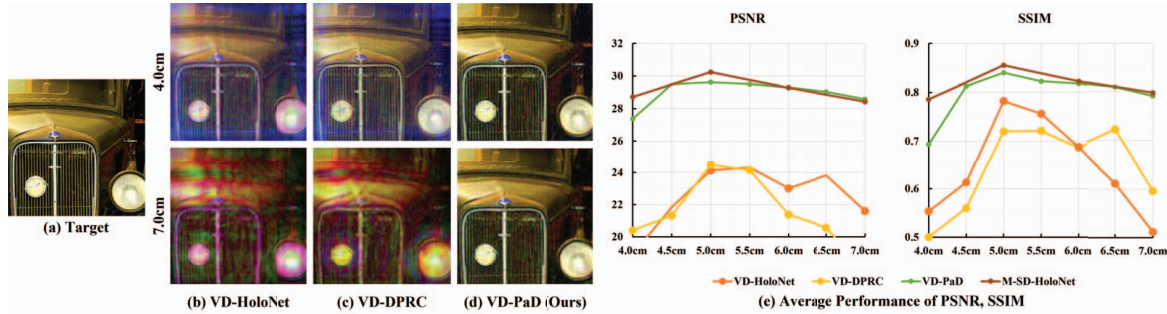


Figure 1: Comparison of numerical reconstruction (NR) performances of three neural hologram generation models trained under variable distances (VD). (b), (c), (d) shows the NR images from the VD-trained HoloNet [3], DPRC [6], and our model, Propagation as Data (PaD), respectively. (e) shows the PSNR and SSIM performance of the three VD-trained models using OpenImage dataset. As can be seen in (b), (c), (d) and (e), the reconstruction performances of HoloNet and DPRC are much lower and unstable, whereas the PaD was able to generate higher-quality phase holograms consistently over all the tested distances.

ABSTRACT

Most of the neural network models for generating phase holograms developed so far are trained and validated only for a single distance. Consequently, if a distance is altered, the performance of models tends to decline dramatically. To address this, we introduce a novel approach called ‘Propagation as Data (PaD)’. Unlike conventional methods, our proposed model does not include the propagation process in a neural network. We pre-calculate propagation kernels and use them as conditioning data. Experimental results demonstrate that our model can consistently generate high-quality phase holograms across a range of distances with a single model.

Index Terms: Computing methodologies—Computer graphics—Image manipulation—Image processing

1 INTRODUCTION

Research on phase hologram generation based on neural networks is being actively conducted, leveraging the fast computation capabilities of GPUs. Specifically, following the success of HoloNet [3], which incorporates the propagation function within the neural network, as illustrated in Figure 2 (a), the advanced researches have been conducted. However, since existing methods are optimized for a single distance, they exhibit severe performance degradation if the distance is changed. Moreover, when the existing models are trained to support variable distance, performances not only degrade but even become unstable across the range of distances, as showcased in Figure 1 (b) and (c). This is due to the propagation function resides in the middle of the neural network. When gener-

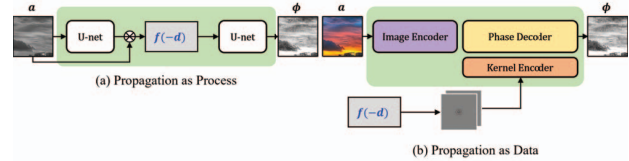


Figure 2: Comparison of phase hologram generation method with a neural network. (a) includes the propagation function within the network, whereas (b) calculates the propagation kernel outside the network.

ating phase holograms with neural networks, the Band-limitation Angular Spectrum Method (BL-ASM) [2] is commonly used to calculate the propagation of light. As the distance between target and hologram plane increases, BL-ASM restricts more frequency band to eliminate the aliasing. This could prevent the network following this propagation function to utilize the whole image information. Since amount of band limitation varies with propagation distances, it can be challenging to train the network to generate proper phase holograms across a range of distances.

To address this issue, we propose a novel method called **Propagation as Data (PaD)**, which can generate phase holograms supporting variable distances with a single model. As shown in Figure 2 (b), we first eliminate the propagation function inside the network to make the whole image information could be utilized. Secondly, we pre-calculate the propagation kernel corresponding to the varying distances and use them as conditioning data for generating phase hologram. As seen in Figure 1 (d) and (e), when trained at variable distances, our PaD model outperforms the conventional methods [3, 6].

2 PROPAGATION AS DATA (PAD)

The model structure, depicted in Figure 3, consists mainly of the image encoder, shallow kernel encoders, and the phase decoder.

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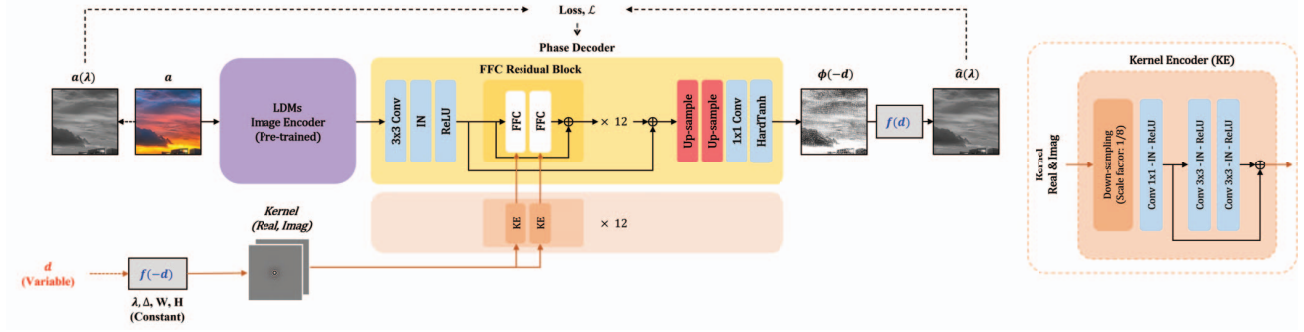


Figure 3: Detail architecture of Propagation as Data (PaD). Image features extracted from the input image a in the pre-trained Image Encoder and the kernel features obtained from the pre-calculated propagation kernel in the Kernel Encoder are integrated in the Phase Decoder to generate phase hologram, $\phi(-d)$.

Image Encoder To train the PaD stably, it is important for image encoder to extract features that well represent the image. We used VQ (Vector Quantization) based image encoder trained on the OpenImage dataset for Latent Diffusion Models (LDMs) [4]. Especially, we made use of the image encoder from the ‘diffusers’ library [5], ‘CompVis/ldm-super-resolution-4x-openimages’. We kept the pre-trained model freeze while training.

Phase Decoder To generate phase hologram, employing global context of image features is crucial for phase decoder. Therefore, we employed Fast Fourier Convolution (FFC) [1] from the early layer of the phase decoder. The FFC module is divided into two main branches: the local branch, which leverages CNN to extract information from localized regions, and the global branch that employs the Fast Fourier Transform (FFT) to transition to the frequency domain, capturing global context information. These branches mutually exchange features through an interaction branch. As depicted in Figure 3, our approach combines two FFC modules with an FFC Residual Block using a skip connection. After 12 FFC Residual Blocks, we can generate phase holograms through the Up-sample module and the HardTanh layer.

Kernel Encoder As depicted in Figure 3, we pre-calculate the propagation kernels based on the given distance, and extract features from the real and imaginary values of those kernels through the kernel encoder. The extracted kernel features and image features are then combined in frequency domain through the global branch of the FFC module. This allows the phase decoder to employ rich image information when generating phase holograms. Moreover, by utilizing the pre-calculated propagation kernels as conditioning data for the given distance, phase decoder can generate appropriate phase holograms corresponding to that specific distance.

3 EXPERIMENTAL RESULTS

In this section, we compare the performance of our model with HoloNet [3], DPRC [6], both trained to support variable distances (VD-HoloNet, VD-DPRC). In addition, multiple HoloNet models optimized for each single distance (M-SD-HoloNet) are also compared. To train the models to support variable distance (VD), each model is trained with distances randomly selected from a uniform distribution from 4cm to 7cm . After training, phase holograms are generated and evaluated at multiple distance with intervals of 0.5cm from 4cm to 7cm . For data augmentation of training dataset, we utilize random crops from the DIV2K dataset, obtaining five 224×224 images from each source image. We train our model using only the Mean Square Error (MSE), whereas other models are trained with their own objective functions as described in their papers.

The results are illustrated in Figure 1. When evaluated on the OpenImage dataset, 6.47dB and 7.53dB average PSNR improvement across multiple propagation distances are observed against VD-HoloNet and VD-DPRC, respectively. The standard deviation of the reconstruction PSNRs across multiple distances is also improved from 1.91 and 2.27 to 0.8, proving the excellence in stability of our model. Furthermore, we observe comparable performances comparing our method to M-SD-HoloNet optimized for each distance. This result can support our conclusion that our PaD approach is suitable for supporting variable distances with a single model.

4 CONCLUSION

We studied the method for generating phase holograms at variable distances using a single model. The proposed PaD approach removes the propagation function in the neural network and uses the propagation kernel as data. Using the proposed method, we were able to generate high-quality phase holograms consistently. As a future research, we can extend our model to support variations not only in propagation distances, but also in the other propagation parameters like wavelength and pixel pitch.

ACKNOWLEDGMENTS

This work was supported by Samsung Research Funding & Incubation Center of Samsung Electronics under Project Number SRFC-IT2201-03 (97%), and Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (No.RS-2022-00155911, Artificial Intelligence Convergence Innovation Human Resources Development (Kyung Hee University)) (3%).

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